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Foreground-aware Universe Graph Matching for Domain Adaptive Object Detection

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ABSTRACT

Domain Adaptive Object Detection (DAOD) focuses on improving the generalization ability of cross-domain object detection via knowledge transfer. Recent advances in DAOD strive to minimize the domain discrepancy for adapting class-conditional distributions by aligning cross-domain sampled node pairs in the non-Euclidean graphical space. However, these methods of reformulating the adaptation with graph matching may fail to concentrate on precise semantics alignment, due to the domain-biased graph modeling and the unreliable matching with background nodes. To solve these issues, a Foreground-aware Universe Graph Matching (FUGM) framework is proposed for DAOD. Specifically, a virtual universe graph is category-wise constructed for modeling semantic knowledge, and node representation is refined via Collaborative Graph Reasoning (CGR), which incorporates with self-loop calibration for the information that node learns about itself and universe graph based semantic interaction. To reduce the spurious matching pair for better semantics alignment, Universe Graph Matching (UGM) is developed to encourage instance nodes to match anchor nodes and remove absorbing nodes. The resulting matching model enables end-to-end learning especially for instance node pairwise affinity enhancement through gathering foreground nodes of corresponding category together. Extensive experiments verify that FUGM outperforms existing works significantly.

1. Introduction

With the tremendous progress in the development of deep learning, object detection has attracted considerable attention in recent years, serving as one of the prevalent visual tasks. Promising performance has been achieved by the generic methods (Carion et al., 2020; Girshick, 2015; Redmon & Farhadi, 2018; Tian, Shen, Chen, & He, 2019) on the assumption with the consistently independent and identically distributed (i.i.d) distribution of training and test data. However, these methods deployed in a novel domain may suffer from catastrophic performance degradation due to the domain gap (Chen, Li, Sakaridis, Dai, & Van Gool, 2018; He, Zhang, Yang, & Gao, 2021; Torralba & Efros, 2011), which seriously restricts the transferability and robustness of object detectors in real-world applications like automatic driving under conspicuously diverse weather conditions and scenes (Arruda et al., 2019; Liu, Zhang, Wan, Ji, & Ye, 2021; Zhang, Zhang, Shi, Fan, & Chen, 2024).

To aim at mitigating the domain discrepancy, Unsupervised Domain Adaptation (UDA) methods have been studied to transfer the object detector from a label-rich domain to a new fully unlabeled domain (Tian, Zhang, Wang, Xiang, & Pan, 2021). Most works focus

on the discriminative representation of each domain and distribution alignment of cross-domain scenes. The pixel-to-pixel adaptation can be achieved by image-level representation and the marginal distribution alignment (Chen, Zheng, Huang, Ding and Yu, 2021; Saito, Ushiku, Harada, & Saenko, 2019). In addition, the instance-level representation has been exploited for the foreground distribution alignment (Hsu, Tsai, Lin, & Yang, 2020; Zhang, Wang and Mao, 2021; Zhu, Liu, Wang, Wang, & Lu, 2023). Recently, semantic-level representation has attracted great attention, serving for class-conditional distribution alignment (Chen, Li et al., 2021; Gao, Liu, Dun, & Qian, 2023; Li, Liu et al., 2023). Among these existing works, significant achievement has been made by semantic-level graph reasoning and distribution alignment in non-Euclidean graphical space (Chen et al., 2022; Li, Liu and Yuan, 2022; Li, Liu, & Yuan, 2023c; Liu, Wang, Huang, Wang and Xu, 2023).

Despite the great success achieved, the aforementioned semantic-level paradigm based on graphs still faces challenges which makes it struggle for robust graph reasoning and matching. The first challenge is that these works rely on the graph reasoning that lacks the modeling of comprehensive semantic space concerning both domains, thereby hindering from narrowing down noisy distributional disparity of the

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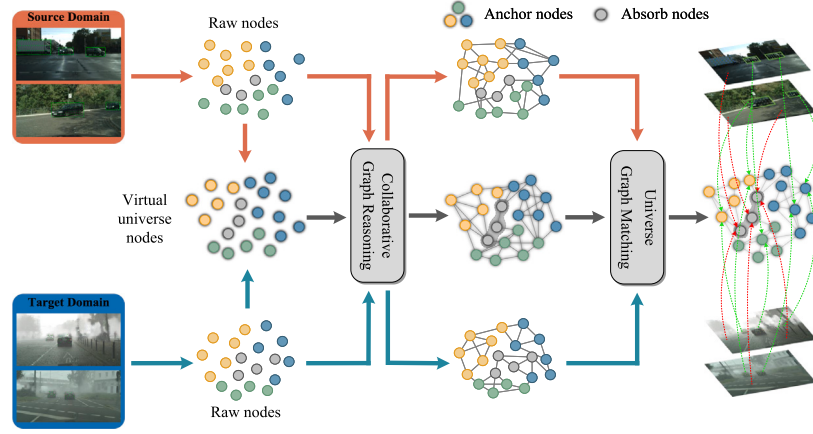


Fig. 1. Illustration of the proposed Foreground-aware Universe Graph Matching (FUGM) framework for domain adaptive object detection.

same class of nodes across domains. For target graph, the noisy nodes may be generated by mislabeled samples from the false pseudo score maps. This further makes it difficult for intra-domain graph model to well characterize the knowledge of inter and intra-class relationships shared by both domains, due to the inherent domain-specific bias (Yuan et al., 2023). Enforcing aligning these cross-domain node embeddings will negatively influence the correct matching and consequently lead to the semantically negative transfer. Therefore, the more promising graph construction beyond a single domain and the corresponding reasoning for capturing universal structural dependencies are greatly needed to robustly represent the cross-domain semantic space.

The second challenge lies in the unreliable matching with background nodes. Semantic-complete node generation for graph matching has been exploited to achieve more effective adaptation (Li, Liu et al., 2022; Li et al., 2023c). The within-class variance of background nodes can be much larger than those of foreground nodes of known categories, due to the higher uncertainty on backgrounds from arbitrary objects of distinct unknown categories (e.g. trees, buildings, roads, etc.). The highly uncertain distribution representation of backgrounds with diverse unknown categories unavoidably brings about the inter-class ambiguity, which may lead to semantic misalignment. There are two kinds of background nodes having spurious matching in bi-graph setting: the nodes of different unknown categories but enforced to match with each other, and the mislabeled nodes matched with known instance nodes by mistakes. The spurious matching particularly between background and other foreground nodes severely damages cross-domain pairwise matching. Hence, it is lack of guidance to recognize node category and reduce the spurious matching pair. As such, the partial matching combined with a comprehensive view (Zhou, Zhu, & Daniilidis, 2015) for node clustering becomes a practical strategy, which gets rid of unreliable matchings and achieves better adaptation.

In this paper, we resolve these challenges from the perspective of robust graph modeling. By exploiting the superior capability of graph representation with capturing structural dependencies for non-Euclidean data (Pei, Wei, Chang, Lei, & Yang, 2020), graph learning produces better reasoning than merely taking the individual node features, which has been successfully applied to natural language processing (NLP) (Ji, Wu, & Lan, 2019; Wu, Chen, Ji, & Liu, 2021) and computer vision (Chen, Wei, Wang and Guo, 2021; Shi & Rajkumar, 2020). Motivated by the intrinsic property of graph modeling that provides the unique advantage in handling instance nodes with relations, we propose a Foreground-aware Universe Graph Matching (FUGM) framework, which fabricates a virtual universe graph for selective matching enhancement. The virtual universe graph is constructed with anchor and absorbing nodes which shares the common semantic space with the real graphs generated from image features in both domains

As shown in Fig. 1, we design a Collaborative Graph Reasoning (CGR) module to model comprehensive knowledge concerning both

domains and the universe graph representation is served as a filter to distinguish node category. The CGR optimizes the universe node representation in semantic space by capturing universal semantic dependencies. Then, to perform robust cross-domain distribution alignment, we develop Universe Graph Matching (UGM) adaptor to cope with background nodes and spurious matchings. Both real graphs are first matched to the universe graph and then pairwise matching is reconstructed from universe affinity. The universe graph provides a matching template that gathers the nodes of the same category and the absorbing nodes are utilized to filter out background nodes. Foreground-aware graph matching loss is further designed to weaken the unreliable matching with background nodes, which encourages learning for affinity measure between foreground instance nodes of identical categories. To be summarized, our main contributions are as follows.

- We propose a Foreground-aware Universe Graph Matching (FUGM) framework for DAOD, which provides a virtual universe graph based partial matching strategy for robust class-conditional distribution alignment.
- We propose Collaborative Graph Reasoning (CGR) module for learning the universe node representation by capturing the knowledge of inter and intra-class relationships shared by both domains. Information aggregation with self-loop calibration and universe graph based interaction are incorporated to robustly model the class-conditional distribution.
- We develop Universe Graph Matching (UGM) adaptor that reformulates DAOD as a partial graph matching problem to bridge the domain gap. The universe affinity learning is utilized for distinguish background nodes and foreground-aware graph matching loss is designed for reducing unreliable matching.
- Comprehensive experiments verify that the propose FUGM achieves state-of-the-art detection performance on three benchmark datasets and outperforms existing counterparts significantly.

2. Related work

2.1. Domain adaptive object detection

Domain adaptive object detection (DAOD) aims to mitigate the cross-domain distributional disparity problem in object detection. Extensive works have been conducted, generally categorizing into four methods. Pixel-level methods (Liu et al., 2021; Saito et al., 2019; Zhao & Wang, 2022) make use of data augmentation and extra interpolated data on the image space for better adaptation. Label-level methods leverage self-training strategies to refine pseudo features (Munir, Khan, Sarfraz, & Ali, 2021; Ramamonjison, Banitalebi-Dehkordi,

Kang, Bai, & Zhang, 2021). Model-level methods (Deng, Li, Chen, & Duan, 2021; He et al., 2022; Li, Dai et al., 2022) rely on an extra teacher model for distillation based learning with cross-domain consistency constraints. Representation-based approaches are prevailing which focus on adversarial learning (Tzeng, Hoffman, Saenko, & Darrell, 2017) and cross-domain distribution alignment for adaptation (Xu et al., 2022). These works follow diverse representation mechanisms to align marginal distribution (Saito et al., 2019) and class-conditional distribution (Chen, Li et al., 2021; Chen et al., 2022; Gao et al., 2023; Li, Liu et al., 2023; Xu, Wang, Ni, Tian, & Zhang, 2020). Some works align the distribution with limited class prototypes from instances which are sub-optimal for aligning the non-convex deep feature distribution (Wang, Luo, Huang, & Baktashmotlagh, 2020; Xu et al., 2020). To alleviate the optimization inconsistency problem between domain alignment and classification, meta learning framework (Zhang et al., 2024; Zhu et al., 2023) have been developed which transfers the meta knowledge across different domains. To align the class-conditional distribution with the essential within-class variance, semantic-level graph-based approaches have made significant achievement (Li, Liu et al., 2022; Li et al., 2023c). However, these works rely on the domain-specific graph based semantic space modeling, which is limited by domain-biased node embeddings to well characterize the knowledge shared by both domains. In this work, we focus on virtual universe graph reasoning to model the comprehensive semantics concerning both domains. Different from foregroundness based method (Liu, Wang, Xiao et al., 2023) incorporating the metric of label quality to improve classification confidence with location accuracy, we leverage universe graph for foreground-aware distribution alignment by seeking for instance-to-anchor node matching.

2.2. Graph learning

Node-to-node edge connections can be modeled by capturing the second-order dependency, which establish a structural representation with geometric knowledge. Aimed at mining the superior capability of non-Euclidean data graph representation, neural graph learning exploits layer-wise knowledge propagation mechanism to aggregate adjacent node features with graph convolution (Pei et al., 2020). Beyond the application on graph data with known edges, graph learning with agnostic structural dependencies has been widely extended in (1) natural language processing (Wu et al., 2021) for visual question answering (Wang, Yasunaga, Ren, Wada, & Leskovec, 2023), and semantic parsing (Ji et al., 2019), etc., and (2) computer vision tasks of image classification (Chen, Wei et al., 2021), scene graph generation (Chang et al., 2021), semantic segmentation (Zhang, Xiao, Jiao, Wei and Zhao, 2021), object detection and tracking (Shi & Rajkumar, 2020; Weng, Wang, Man, & Kitani, 2020), etc. Existing efforts on DAOD establish the learnable structural edges of input graphs without explicit edge annotations and refine node representation to complete the mismatched semantics for domain adaptation (Li, Liu et al., 2022; Li et al., 2023c). However, these works overlook the domain-biased edge learning caused by the domain shift. Instead of independent language graph learning (Liu, Wang, Huang et al., 2023) which may bring about the cross-modal gap in universal structural edge learning, we leverage virtual universe graph for collaborative learning from comprehensive view.

2.3. Graph matching

Graph matching learns the pair-wise node correspondences between two graphs, which establishes the one-to-one matching of cross-graph nodes (Gao, Wang, Xue, Yu, & Xia, 2021). Graph matching problem can be formulated as a quadratic assignment problem (QAP) (Loiola, De Abreu, Boaventura-Netto, Hahn, & Querido, 2007), which is well-known NP-complete (Hartmanis, 1982). The matching solvers (Yan

Table 1

Comparisons of graph-based methods.

Method	Graph reasoning	Graph matching
SIGMA (Li, Liu et al., 2022)	Shared in source/target domain graphs	Full matching
SIGMA++ (Li et al., 2023c)	Shared in source/target domain graphs	Full matching
CIGAR (Liu, Wang, Huang et al., 2023)	Shared in source/target domain graphs but independent in language	Full matching
FUGM (ours)	Collaborative in source/target domain graphs and universe graph	Partial matching

et al., 2016) optimize a cross-graph assignment matrix or correspondence matrix to encode both node and edge affinities. Graph matching has been extended to visual correspondence (Yang, Liu, & Qiao, 2020), point cloud registration (Fu et al., 2022) and DAOD (Li, Liu et al., 2022; Liu, Wang, Huang et al., 2023) to model pair-wise relationships in the graphical space. Yang et al. (2020) model the visual feature correspondence problem by establishing graph matching between pairwise key point features on images covering the same objects. Fu et al. (2022) solve the point cloud registration problem by establishing graph matching between pairwise point features on point homogeneous sets. Recently, the class-distribution alignment in DAOD has been reformulated as a node pairwise graph matching problem and the QAP solver has been adopted bridging the domain gap (Li, He et al., 2023; Li, Liu et al., 2022; Li et al., 2023c). While these methods allow end-to-end learning, they focus on the unrealistic setting of full matchings, where there are spurious correspondences with respect to background nodes. As is shown in Table 1, we compare the existing graph-based methods for DAOD. Different from fairly treating node pairwise matchings in cross-modality linguistic prototype-guided graph matching (Liu, Wang, Huang et al., 2023), we propose to seek an instance-to-anchor matching with partial universe affinity learning, where within-class variance is also modeled in universe subgraph.

3. Proposed methodology

The overall workflow of the proposed FUGM framework is shown in Fig. 2. Consider batch-wise annotated source images $D_s = \{(x_s^i, y_s^i)\}_{i=1}^B$ and unlabeled target images $D_t = \{x_t^i\}_{i=1}^B$ with K known categories. Our approach begins by leveraging a shared feature extractor ϕ to obtain visual features $F_{s/t} = \{\phi(x_{s/t}^i)\}_{i=1}^B \in \mathbb{R}^{h \times w \times d}$ where (h, w) and d are the size and channel dimension of a feature map. The DAOD is aimed at transferring knowledge from D_s to D_t for improving the detection performance in D_t , where the data in the distinct domains are non-identically distributed. We consider graph modeling for both of instances and backgrounds from the uniformly sampled visual features. The nodes of missing classes can be generated from a memory bank (Vs, Gupta, Oza, Sindagi, & Patel, 2021) for obtaining semantic-complete raw node sets $\mathcal{V}_{s/t}$ with identical $K + 1$ classes of K known categories for instances and one unknown category for background (Li, Liu et al., 2022). Meanwhile, the virtual universe node set $\mathcal{V}_u$ is category-wise constructed for modeling comprehensive semantics of both domain through Universe Graph Generator (UGG).

Then, the three sets are feed into a Collaborative Graph Reasoning (CGR) module. Intra-graph neighborhood information is aggregated by Collaborative Intra-Graph Reasoning (CIGR) with improved graph convolution. The cross-image graphs $\bar{\mathcal{G}}_{s/t}$ with universe graph $\bar{\mathcal{G}}_u$ are established to model class-conditional distribution with enhanced nodes. CIGR is followed by Collaborative Cross-Graph Reasoning (CCGR), where cross-graph knowledge reasoning to accomplish universe graph based knowledge interaction with cross-image graphs. The anchor and absorbing nodes of the virtual universe graph $\hat{\mathcal{G}}_u$ are well constructed for foreground-aware node affinity learning. Afterward, the refined



concerning both domains and optimize the universe node representation in semantic space by capturing cross-image and cross-graph semantic structural dependencies for $\mathcal{G}_{s/t/u} = \{\mathcal{V}_{s/t/u}, \mathcal{E}_{s/t/u}\}$ with edge set $\mathcal{E}_{s/t/u}$. Our collaborative learning procedure involves the message propagation within each graph and across graphs. In this section, we propose intra-graph reasoning with self-loop calibration (SLC) and source/target-to-universe cross-graph reasoning to optimize the inter-class and intra-class distribution representation by capturing universal structural dependencies.

3.2.1. Collaborative intra-graph reasoning

The improved graph convolution network is introduced to perform Collaborative Intra-Graph Reasoning (CIGR). Node affinity is learned with self-attention to model the class-conditional distribution with long-distance semantic dependency. The information aggregation of neighbors can be well realized under different node relationships, and the self-loop is dynamically calibrated so that it can well maintain the original features. The improved graph convolution layer is followed by layer normalization and a residual connection:

$$\tilde{v}_{s/t/u} = v_{s/t/u} + \text{LN}(\tilde{v}_{s/t/u}), \quad (2)$$

where denotes the layer normalization, the node representation $\tilde{v}_{s/t/u}$ is obtained by improved graph convolution with the message propagation among nodes to aggregate intra-graph semantic knowledge:

$$\tilde{v}_{s/t/u}^i = \sum_{v_{s/t/u}^j \in \delta(i)} \mathcal{A}_{s/t/u}^{i,j} v_{s/t/u}^j \mathcal{W}_{igc} + (1 + \epsilon) v_{s/t/u}^i, \quad (3)$$

where $\delta(i)$ stands for the neighbor nodes of $v_{s/t/u}^i$, $\mathcal{W}_{igc}$ and ϵ represents learnable matrix and parameter, respectively, $\mathcal{A}_{s/t/u} = \text{softmax}(\mathcal{V}_{s/t/u} \mathcal{W}_{eq} (\mathcal{V}_{s/t/u} \mathcal{W}_{ek})^T)$ is the adjacency matrix encoding structure information learned with self-attention, $\mathcal{W}_{eq}$ and $\mathcal{W}_{ek}$ denote learnable linear projections. ϵ is regarded as the learnable SLC weight of the remained information of original node. Compared with semantic prototype based graph reasoning (Liu, Wang, Huang et al., 2023), the CIGR is based on universe graph with richer intra-graph information concerning essential within-class modeling, and it aggregates similar neighboring information as well as adaptively preserves the original feature for $\mathcal{G}_{s/t/u} = \{\mathcal{V}_{s/t/u}, \mathcal{E}_{s/t/u}\}$. According to (3), the learned adjacency matrix $\mathcal{A}_{s/t/u}$ encodes semantic affinity between nodes across domains, which could capture the domain-invariant semantic relationship using the attention within CIGR. The each entry of the matrix stands for the strength of connection between node pairs, which can provide the reliable structure information used for graph matching.

3.2.2. Collaborative cross-graph reasoning

The universe graph is utilized to establish cross-graph attention, which is incorporated to perform Collaborative Cross-Graph Reasoning (CCGR). The message propagation across graphs is essential for the optimal solution in graph-based affinity learning. Inspired by Li, Liu et al. (2022), the node representations can be improved by the collaborative reasoning between the source/target and universe graphs.

$$\begin{aligned} \hat{v}_{s/t} &= \text{LN}(\bar{v}_{s/t} + \text{MLP}(\mathcal{A}(\bar{v}_{s/t}, \bar{v}_u) \bar{v}_u)), \\ \hat{v}_u &= \text{MLP}(\text{concat}(\bar{v}_{su}, \bar{v}_{tu})), \\ \bar{v}_{su/tu} &= \text{LN}(\bar{v}_u + \text{MLP}(\mathcal{A}(\bar{v}_u, \bar{v}_{s/t}) \bar{v}_{s/t})), \end{aligned} \quad (4)$$

where cross-graph attention can be obtained as:

$$\begin{aligned} \mathcal{A}(\bar{v}_{s/t}, \bar{v}_u) &= \text{softmax}(\bar{v}_{s/t} \bar{\mathcal{W}}_q (\bar{v}_u \bar{\mathcal{W}}_k)^T), \\ \mathcal{A}(\bar{v}_u, \bar{v}_{s/t}) &= \text{softmax}(\bar{v}_u \bar{\mathcal{W}}_q (\bar{v}_{s/t} \bar{\mathcal{W}}_k)^T), \end{aligned} \quad (5)$$

where $\bar{\mathcal{W}}_q$ and $\bar{\mathcal{W}}_k$ represents learnable parameters. Thus, CCGR encodes the cross-graph class-conditional distribution with long-distance

universal semantic dependencies, which promotes debiased collaborative edge learning. The universe graph based semantic interaction encourages the correspondence of within-class nodes from universe and source/target graph, which facilitates fine-grained adaptation with interactive semantic cues. To enhance the node semantics of cross-image graphs (Li, Liu et al., 2022), a classifier f_{cls} with the cross-entropy loss is adopted in auxiliary node classification task of CCGR:

$$\mathcal{L}_{NC} = - \sum_{i=1}^{N_s+N_t} y_{s/t}^i \log \left\{ \text{softmax}[f_{cls}(\hat{v}_{s/t}^i)] \right\}, \quad (6)$$

where $y_{s/t}^i$ represents the ground-truth labels for source nodes and the pseudo labels for target nodes. This guarantees the domain-specific semantic consistency within CGR framework.

3.3. Universe graph matching

Given the CGR enhanced graph nodes, the cross-domain distribution alignment in DAOD has been reformulated as a graph matching problem. Cross-graph node affinity is learned with cross-attention to model the universal cross-category relationship as well as essential within-class difference. Existing graph matching approaches focus on alignment of the visual features at the category level, leveraging a pairwise affinity matrix for the node matching setting (Fu et al., 2022; Wang, Yan and Yang, 2019):

$$S_{st} = \hat{v}_s \mathcal{W}_{st} \hat{v}_t^T, \quad (7)$$

where $\mathcal{W}_{st}$ is a learnable weight of the node affinity. Each entry of the pairwise affinity can be regarded as the class-wise transition probability between source and target nodes. As for full matching, the uncertain outliers like background nodes of unknown category will destroy the semantic relationship across domains, since these background nodes are excluded from each other and randomly scattered in the feature space in previous work. It is possible to result in cross-domain category-level confusion and degrading the quality of the pairwise affinity function. Therefore, it is necessary for considering the partial matching that is robust against the spurious correspondence where background nodes are regarded as uncertain outliers.

3.3.1. Universe Node Affinity (UNA)

To overcome the challenges brought by full matching, we propose universe graph matching (UGM) method for partial matching. Specifically, we fabricate the universe graph $\hat{v}_u$ with anchor nodes from the subgraphs $\{\hat{v}_u^k\}_{k=1}^K$ and absorbing nodes from the $(K+1)$ th subgraph $\hat{v}_u^{K+1}$. The universe anchor features separate the latent feature space into different semantic parts that encourages the similar node features to be matched, while the absorbing nodes are expected to get close to the background nodes.

The universe affinity $S_{su/tu} \in \mathbb{R}^{N_{s/t} \times N_u}$ between source/target graph and the universe graph can be learned:

$$S_{su/tu} = \hat{v}_{s/t} \mathcal{W}_{su/tu} \hat{v}_u^T = \hat{v}_{s/t} F_{su/tu}, \quad (8)$$

where $\mathcal{W}_{su/tu}$ and $\hat{v}_u^T$ share weight for all matching instance, $F_{su/tu} = \mathcal{W}_{su/tu} \hat{v}_u^T$ can be regarded as learnable weight. This is equivalent to performing a linear transformation on the features. We also add softmax to increase the non-linearity in metric learning, which can extract universe affinity of higher quality.

Then the universe affinity is utilized to reconstruct the pairwise affinity. Prior to this, the learnable weight $F_{su/tu}$ is adopted to construct the learnable weight of the node affinity in (7):

$$\mathcal{W}_{st} = F_{su/tu} F_{su/tu}^T \quad (9)$$

By substituting (9) into (7), the pairwise affinity matrix for the cross-domain node matching can be reformulated as:

$$S_{st} = \hat{v}_s F_{su} F_{tu}^T \hat{v}_t^T. \quad (10)$$

Table 2

The results on Cityscapes $\rightarrow$ Foggy Cityscapes (%) using VGG-16 and ResNet-50 backbone networks with mAP as the evaluation metric. SO represents the results obtained with only training on the source domain, and GAIN represents the adaptation gains compared to the source-only (SO) results.

Method	Backbone	Person	Rider	Car	Truck	Bus	Train	Motor	Bike	mAP	SO	GAIN
TIA (Zhao & Wang, 2022)	VGG-16	34.8	46.3	49.7	31.1	52.1	48.6	37.7	38.1	42.3	20.3	22.0
MGA (Zhou, Du, Zhang, Luo, & Wu, 2022)		43.9	49.6	60.6	29.6	50.7	39.0	38.3	42.8	44.3	25.2	18.8
FGRR (Chen et al., 2022)		33.5	46.4	49.7	28.2	45.9	39.7	34.8	38.3	39.6	19.3	20.8
TDD (He et al., 2022)		39.6	47.5	55.7	33.8	47.6	42.1	37.0	41.4	43.1	17.3	25.8
SSAL (Munir et al., 2021)		45.1	47.4	59.4	24.5	50.0	25.7	26.0	38.7	39.6	20.4	19.2
EPM (Hsu et al., 2020)		41.9	38.7	56.7	22.6	41.5	26.8	24.6	35.5	36.0	18.4	17.6
SCAN++ (Li, Liu et al., 2023)		44.2	43.9	57.9	28.2	48.1	51.2	30.1	39.5	42.8	18.4	24.4
SIGMA (Li, Liu et al., 2022)		46.9	48.4	63.7	27.1	50.7	35.9	34.7	41.4	43.5	18.4	25.1
SIGMA++ (Li et al., 2023c)		46.4	45.1	61.0	32.1	52.2	44.6	34.8	39.9	44.5	18.4	26.1
OADA (Yoo, Chung, & Kwak, 2022)		47.3	45.6	62.8	30.7	48.0	49.4	34.6	39.5	44.8	–	–
CIGAR (Liu, Wang, Huang et al., 2023)		45.3	45.3	61.6	32.1	50.0	51.0	31.9	40.4	44.7	18.4	26.3
AWADA (Menke, Wenzel, & Schwung, 2024)		44.7	52.7	60.9	28.7	51.0	36.2	36.0	48.5	44.8	–	–
VIDSN (Chen et al., 2023)		45.0	55.2	61.7	29.3	44.0	29.0	36.2	46.9	43.1	–	–
MRDA (Bai, Liu, Yang, & Li, 2024)		48.7	50.1	64.1	25.6	46.0	50.4	34.3	38.8	44.8	18.4	26.4
OMLF (Zhu et al., 2023)		38.6	46.3	57.9	29.6	49.9	50.8	32.9	40.0	42.6	20.3	22.3
GRDIV (Xi, Liu, Lu, Ding, & Ding, 2025)		47.0	47.9	62.1	29.5	52.8	53.2	34.7	41.9	45.8	18.1	26.8
CRADA (Liu et al., 2024)		47.0	48.3	64.1	28.7	52.3	38.6	35.1	41.5	44.5	18.4	26.1
FUGM (ours)		45.5	44.4	61.4	33.2	54.3	54.5	34.9	40.1	46.0	18.4	27.6
GPA (Xu et al., 2020)	ResNet-50	32.9	46.7	54.1	24.7	45.7	41.1	32.4	38.7	37.4	22.8	16.7
DIDN (Lin et al., 2021)		38.3	44.4	54.8	28.7	53.3	34.7	32.4	40.4	40.5	28.6	11.9
DSS (Wang et al., 2021)		42.9	51.2	53.6	33.6	49.2	18.9	36.2	41.8	40.9	22.8	18.1
SDA (Rezaeianaran et al., 2021)		38.8	45.9	57.2	29.9	50.2	51.9	31.9	40.9	43.3	22.8	20.8
EPM (Hsu et al., 2020)		39.9	38.1	57.3	28.7	50.7	37.2	30.2	34.2	39.5	24.2	15.3
SIGMA (Li, Liu et al., 2022)		44.0	43.5	60.3	31.6	50.4	51.5	31.7	40.6	44.2	24.2	20.0
CIGAR (Liu, Wang, Huang et al., 2023)		46.1	47.3	62.1	27.8	56.6	44.3	33.1	41.3	44.9	24.2	20.7
OMLF (Zhu et al., 2023)		38.8	45.9	57.2	29.9	50.2	51.9	31.9	40.9	43.3	22.8	20.5
CRADA (Xi et al., 2025)		45.6	43.8	61.8	30.7	51.0	52.1	34.2	38.9	44.8	24.2	20.6
VLDadaptor (Ke et al., 2024)		44.7	43.9	62.0	33.8	19.8	55.7	34.6	39.0	45.6	24.2	21.4
FUGM (ours)		44.4	45.3	61.1	34.2	53.8	57.8	33.9	38.8	46.2	24.2	22.0

Then, we substitute (8) into (10). The pairwise affinity could be further reconstructed by universe affinity $S_{su/tu}$ as follows:

$$S_{st} = S_{su} S_{tu}^T. \quad (11)$$

According to (11), the pairwise affinity could be reconstructed by universe affinity without information loss. Compared with each graph from the single domain, universe graph aggregates the information of both domains which could be served as a bridge for matching across domains.

Since background nodes are randomly scattered in the feature space, directly full matching may suffer from the spurious correspondence between the instance nodes and the backgrounds. For the robustness against the spurious correspondence, the filtered universe affinity is exploited to achieve instance-to-anchor matching. The absorbing node removal (ANR) technique can be exploited to filter out background nodes by removing the last several columns of universe affinity $S_{su/tu}$:

$$\hat{S}_{su/tu} = S_{su/tu} \odot \mathcal{M}, \quad (12)$$

where $\odot$ denotes Hadamard product and $\mathcal{M} \in \mathbb{R}^{N_{s/t} \times N_u}$ denotes the selection matrix. In this work, the elements of last $N_u^{bg} = \max(N_s^{K+1}, N_t^{K+1})$ columns are ones with others being zeros, where N_u^{bg} denotes number of absorbing nodes in universal graph. The final cross-domain pairwise affinity matrix for partial matching can be obtained as:

$$\hat{S}_{st} = \hat{S}_{su} \hat{S}_{tu}^T \in \mathbb{R}^{N_s \times N_t} \quad (13)$$

where $\hat{S}_{su/tu}$ can be partially learned with the affinity between background and absorbing nodes never reduced and the instance-to-anchor correspondences enhanced. As a result, the background nodes are clustered into the absorbing node even without the guidance of the ground truth matching label, and they can be distinguished from instance nodes which allows for effective learning of instance features.

3.3.2. Universe graph matching loss

We learn the universe affinity matrix $\hat{S}_{st}$ for cross-domain nodewise alignment. The element $\hat{S}_{st}^{i,j}$ represents the homogeneous affinity between $\hat{v}_s^i \in \hat{\mathcal{V}}_s$ and $\hat{v}_t^j \in \hat{\mathcal{V}}_t$. Different from structure-aware loss for

cross-domain full node-to-node graph matching (Li, Liu et al., 2022), we provide structure-aware universe graph matching loss by incorporating the universe affinity matrix with instance-to-anchor matching enhancement through ANR:

$$\begin{aligned} \mathcal{L}_{UGM} = & \sum_i N_s^{-1} [\max_j (\hat{S}_{st} \odot \mathbf{Y}_{st})_{i,j} - 1]^2 \\ & + \sum_{i,j} \|1 - \mathbf{Y}_{st}\|_1^{-1} [\hat{S}_{st} \odot (1 - \mathbf{Y}_{st})]_{i,j}^2 \\ & + \sum_{i,j} (N_s N_t)^{-1} (\mathcal{A}_s \hat{S}_{st} - \hat{S}_{st} \mathcal{A}_t)_{i,j}, \end{aligned} \quad (14)$$

where $\mathbf{Y}_{st}$ is the matching label matrix and $Y_{st}^{i,j} = 1$, if nodes $v_s^i \in G_s$ and $v_t^j \in G_t$ belong to the same category; otherwise, $Y_{st}^{i,j} = 0$. The first two terms in the equation promotes the model to match pair-wise nodes correctly, thus aligning foreground instance node features at the category level. The third term optimizes the model to narrow down structural difference of matched node pairs, thus capturing universal structural dependencies.

3.4. Overall loss

At the training stage, we adopt class-agnostic global alignment on visual features with global adversarial (GA) loss $\mathcal{L}_{GA}$ (Hsu et al., 2020). Considering the non-grid matching relationship among graph nodes in the non-euclidean graphical space, we adopt a domain discriminator (three stacked FC-LayerNorm-ReLU blocks) to align the matching node pairs and mitigate the domain gap in the graphical space with the domain adversarial (DA) loss $\mathcal{L}_{DA}$ (Ganin et al., 2016). The overall loss function of the model consists of the following five components:

$$\mathcal{L} = \mathcal{L}_{det} + \mathcal{L}_{GA} + \mathcal{L}_{DA} + \lambda_1 \mathcal{L}_{UGM} + \lambda_2 \mathcal{L}_{NC}, \quad (15)$$

where $\mathcal{L}_{det}$ represents the FCOS object detection loss (Tian et al., 2019), $\mathcal{L}_{GA}$ is the global adversarial loss for global image domain alignment, $\mathcal{L}_{DA}$ is the domain adversarial loss for graph node domain alignment, $\mathcal{L}_{UGM}$ is the universe graph matching loss, $\mathcal{L}_{NC}$ is the auxiliary node classification loss, λ_1 and λ_2 are hyperparameters.

Table 3
Comparison results (%) on Sim10K $\rightarrow$ Cityscapes (S $\rightarrow$ C) and KITTI $\rightarrow$ Cityscapes (K $\rightarrow$ C) with VGG-16 backbone.

Method	S $\rightarrow$ C	SO/GAIN	K $\rightarrow$ C	SO/GAIN
EPM (Hsu et al., 2020)	49.0	39.8/9.2	43.2	34.4/8.8
DSS (Wang et al., 2021)	44.5	34.7/9.8	42.7	34.6/8.1
SSAL (Munir et al., 2021)	51.8	38.0/13.8	45.6	34.9/10.7
SCAN++ (Li, Liu et al., 2023)	53.1	39.8/13.3	46.4	34.4/12.0
SIGMA (Li, Liu et al., 2022)	53.7	39.8/13.9	45.8	34.4/11.4
SIGMA++ (Li et al., 2023c)	57.7	39.8/17.9	49.5	34.4/15.1
OADA (Yoo et al., 2022)	56.6	–	46.3	–
CSDA (Gao et al., 2023)	57.8	–	48.6	–
CIGAR (Liu, Wang, Huang et al., 2023)	58.5	39.8/18.7	48.5	34.4/14.1
AWADA (Menke et al., 2024)	54.1	–	–	–
VIDSN (Chen et al., 2023)	53.5	–	–	–
MRDA (Bai et al., 2024)	55.2	39.8/15.4	–	–
OMLF (Zhu et al., 2023)	46.5	34.3/12.2	46.3	34.6/11.7
CRADA (Xi et al., 2025)	57.6	39.8/17.8	48.1	34.4/13.7
FUGM (ours)	59.0	39.8/19.2	49.8	34.4/15.4

4. Experimental results

4.1. Datasets and evaluation

To validate our proposed method, we conduct extensive experiments on three different domain adaptation object detection datasets (Chen et al., 2018; Hsu et al., 2020; Li, Liu et al., 2022). We adopt FCOS (Tian et al., 2019) as the base detector and employed mean Average Precision with a threshold of 50% (mAP) as the evaluation metric. Additionally, we compare the source-only (SO) results of the model trained on the source domain and tested on the target domain, calculating the adaptation gain (GAIN) which indicates the detection gains compared with SO trained models.

4.1.1. Cityscapes $\rightarrow$ Foggy Cityscapes

Cityscapes (Cordts et al., 2016) is a dataset that consists of real urban scenes under clear weather conditions. There are 2975 annotated images of eight classes in the training set and 500 images in the validation set. Foggy Cityscapes (Sakaridis, Dai, & Van Gool, 2018) is a synthesized dataset generated from the Cityscapes dataset under heavy foggy weather conditions. On these datasets, we investigate the weather-based domain adaptation performance from clear to foggy weather.

4.1.2. Sim10k $\rightarrow$ Cityscapes

Sim10K (Johnson-Roberson et al., 2017) is a simulated dataset sampled from Grand Theft Auto V, which is composed of 10,000 images containing only bounding boxes of the car category. There is inherent domain gap between simulated and real datasets sampled from the real world scene (Cityscapes). We aim to evaluate the domain adaptation performance from synthesized to real-world data.

4.1.3. KITTI $\rightarrow$ Cityscapes

KITTI (Geiger, Lenz, & Urtasun, 2012) is similar to Cityscapes, which is a real-world scene dataset collected from different vehicle-mounted cameras. This dataset comprises 7481 images containing only the car category. We utilize these two datasets to evaluate the cross-camera domain adaptation performance.

4.2. Implementation details

We utilize VGG-16 (Simonyan & Zisserman, 2015) and ResNet-50 (He, Zhang, Ren, & Sun, 2016) as backbone networks to extract visual features. Stochastic Gradient Descent (SGD) optimizer (Paszke et al., 2019) is chosen as the model optimizer with a learning rate of 0.0025, momentum of 0.9, and weight decay of 5×10^{-4} . The batch size is set to 4. The thresholds τ_{fg} and τ_{bg} are empirically set to 0.5 and 0.05, respectively, which are commonly used in existing object detectors (Tian et al., 2019). In the loss function, the hyperparameters λ_1 and λ_2 are set to 0.01 and 0.1, respectively. All experiments were conducted on a single NVIDIA TITAN RTX GPU.

Table 4

Ablation studies on Cityscapes $\rightarrow$ Foggy Cityscapes (C $\rightarrow$ F) and KITTI $\rightarrow$ Cityscapes (K $\rightarrow$ C) with VGG-16 backbone.

Method	w/o	C $\rightarrow$ F			K $\rightarrow$ C		
		$AP_{0.5:0.95}$	$AP_{0.5}$	$AP_{0.75}$	$AP_{0.5:0.95}$	$AP_{0.5}$	$AP_{0.75}$
Baseline	–	17.0	35.3	15.4	–	43.2	–
+CGR	CIGR	23.2	43.2	22	24.4	48.8	22.5
	SLC	23.7	43.1	21.4	23.7	48.8	20.1
	CCGR	21.2	42.0	19.6	25.1	48.5	23.2
	–	23.2	43.6	21.8	24.5	48.9	22.3
+CGR+UGM	UNA	23.0	43.8	21.2	24.2	49.5	21.2
	ANR	23.7	44.8	22.3	23.8	49.6	20.9
	SGF	24.4	45.1	23.5	23.8	49.5	20.7
	–	24.1	46.0	22.4	25.4	49.8	23.8

4.3. Comparison with state-of-the-arts

4.3.1. Cityscapes $\rightarrow$ Foggy Cityscapes

As shown in Table 2, we compare FUGM with previous methods using VGG-16 (Simonyan & Zisserman, 2015) and ResNet-50 (He et al., 2016) as backbones. FUGM achieves 46.0% and 46.2% mAP, respectively, outperforming existing methods. Compared with latest graph matching-based models such as SCAN++ (Li, Liu et al., 2023) (42.8%), SIGMA (Li, Liu et al., 2022) (43.5%), SIGMA++ (Li et al., 2023c) (44.5%) and CIGAR (Liu, Wang, Huang et al., 2023) (44.7%), FUGM achieves of 3.2%, 2.5%, 1.5% and 1.3% mAP improvements with VGG-16 backbone, which validates its advantages over existing graph or hypergraph matching-based methods. Besides, FUGM surpasses recently proposed MRDA (Bai et al., 2024), adaptive graph enhancement adversarial learning based method GRDIV (Xi et al., 2025), hybrid graph adversarial learning based method CRADA (Liu et al., 2024) with 1.2%, 0.2% and 1.5% mAP. This demonstrates that FUGM performs better than the SOTA methods.

4.3.2. Sim10K $\rightarrow$ Cityscapes

The Table 3 lists that our model achieved a performance of 59.0% mAP when using VGG-16 as the backbone, outperforming other methods. Compared to similarly graph matching based methods such as SIGMA (Li, Liu et al., 2022) (53.7%), CIGAR (Liu, Wang, Huang et al., 2023) (58.5%) and recently proposed CRADA (Liu et al., 2024), our method achieves improvements of 5.3%, 0.5% and 1.4% mAP, respectively. This demonstrates that our model still performs well on single-category datasets.

4.3.3. KITTI $\rightarrow$ Cityscapes

The Table 3 lists the performance comparison with VGG-16 as the backbone. Our model achieved a performance of 49.8% mAP, ranking as the best performance among all models outperforming other graph matching or hypergraph matching models such as SCAN++ (Li, Liu

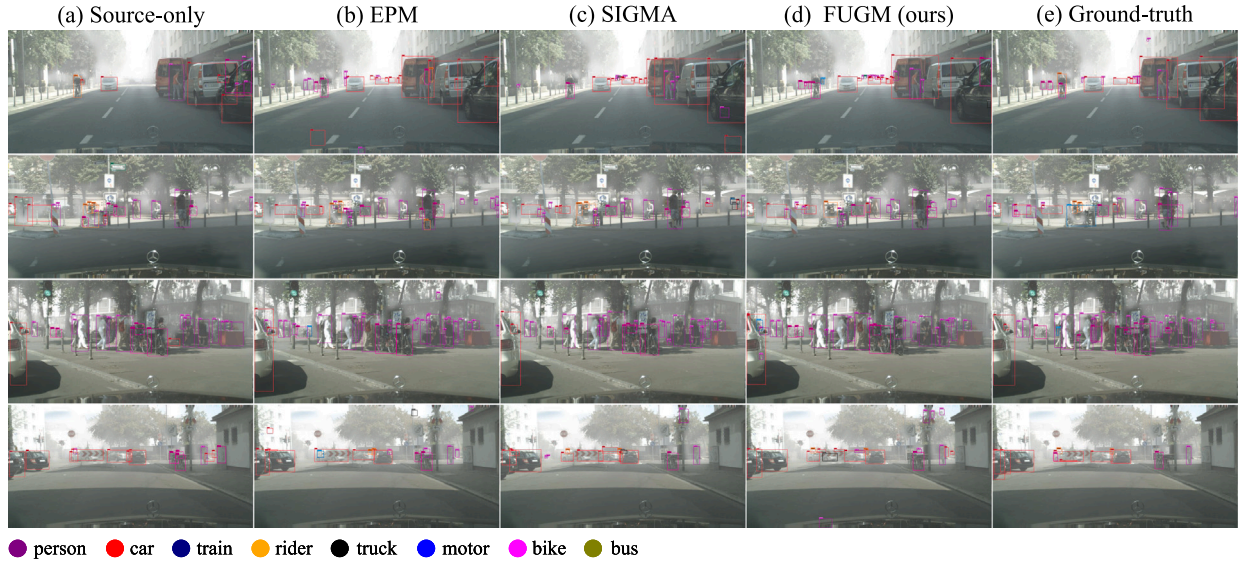


Fig. 3. Result comparison on the Cityscapes → Foggy Cityscapes adaptation scenario among (a) the source only model, (b) EPM (Hsu et al., 2020), (c) SIGMA (Li, Liu et al., 2022), (d) the proposed FUGM and (e) Ground-truth. (Zooming in for best view.)

et al., 2023) (53.1%), CIGAR (Liu, Wang, Huang et al., 2023) (48.5%) and SIGMA++ (Li et al., 2023c) (49.5%). This validates the advantage of our method in terms of adaptation.

4.4. Ablation studies

We present detailed ablation studies as shown in Table 4.

4.4.1. Ablation on the collaborative graph reasoning

As shown in Table 4, adopting only the CGR module can achieve 43.6% and 48.9% mAP on the C → F and K → C datasets, respectively, which improve by 8.3% mAP and 5.7% compared with the GA baseline (Hsu et al., 2020). Subsequently, we sequentially remove each sub-component to verify their effectiveness. Removing the CIGR module weakens intra-graph information aggregation, resulting in 43.2% and 48.8% mAP, respectively. Removing the SLC gives 0.5% mAP and 0.1% mAP drops, indicating that adaptively preserving the original feature is helpful in information aggregation. Removing the CCGR module gives significant drops (1.6% and 0.4% mAP), due to the poor modeling sharing semantic knowledge of universe graph without cross-graph interaction.

4.4.2. Ablation on universe graph matching

As shown in Table 4, incorporating UGM into the framework achieves consistent improvement with a remarkable 46.0% mAP and 49.8% mAP, respectively, which outperforms the GA baseline (Hsu et al., 2020) with 10.7% and 6.6% mAP. Replacing UNA with the pairwise affinity in (7) (Fu et al., 2022; Wang, Yan et al., 2019) gives 2.2% and 0.3% mAP drops due to the lack of a template graph for matching. Removing ANR gives 1.2% and 0.2% mAP drops due to the impact of unreliable matching. Removing SGF gives 0.9% and 0.3% mAP drops due to the lack of the latent feature space separation for universe graph. These demonstrate the effectiveness of UNR, ANR and SGF on cross-domain alignment.

4.5. Sensitivity analysis

In order to better validate and understand our method, we conduct studies on the role of universe graph in CGR and different settings of ANR.

Table 5

Results on Cityscapes → Foggy Cityscapes (C → F) and Sim10K → Cityscapes (S → C) w.r.t. universe graph in CGR.

Method	CIGR	CCGR	C → F			S → C		
			$AP_{0.5:0.95}$	$AP_{0.5}$	$AP_{0.75}$	$AP_{0.5:0.95}$	$AP_{0.5}$	$AP_{0.75}$
Universe graph	✓		23.1	43.2	21.7	34.1	57.8	33.5
			22.9	43.3	21.5	34.6	58.1	35.0
	✓	✓	24.0	44.2	22.3	34.6	58.6	35.3
			24.1	46.0	22.4	35.8	59.0	36.0

Table 6

Results on Cityscapes → Foggy Cityscapes (%) with different settings of ANR.

Method	Person	Rider	Car	Truck	Bus	Train	Motor	Bike	mAP
1 remove	46.5	46.2	62.6	29.5	54.8	43.3	34.0	41.0	44.8
50% drop	46.7	44.6	62.5	31.9	46.9	56.1	33.5	39.8	45.2
Ours	45.5	44.4	61.4	33.2	54.3	54.5	34.9	40.1	46.0

4.5.1. Evaluation on the role of universe graph in CGR

As is shown in Table 5, the universe graph in both of CIGR and CCGR can enhance the detection performance. This is because the universe graph is helpful in both collaborative graph reasonings to improve the semantic-level representations for the best performance of our FUGM. In the first row, CGR is replaced by intra-domain graph reasoning without universe graph, giving a significant drop (43.2% mAP). This indicates the initialized universe graph directly utilized for UGM lead to the mismatching.

4.5.2. Evaluation on the ANR settings

Table 6 shows the effectiveness of the ANR. Compared with a single absorbing node removal and 50% absorbing node removal, our ANR gives 1.2% and 0.8% mAP improvements. “1 remove” and “50% drop” imply the elements of selection matrices of the last column and the 50% last N_u^{bg} columns are ones with the others being zeros, respectively. These results demonstrate that absorbing nodes are served as the universal templates that are matched with background nodes, and they can be removed for better foreground-aware graph matching.

4.6. Qualitative results

4.6.1. Object detection visualization

We present the object detection visualization results of (a) EPM (Hsu et al., 2020), (b) SIGMA (Li, Liu et al., 2022), (c) ours, and (d) ground

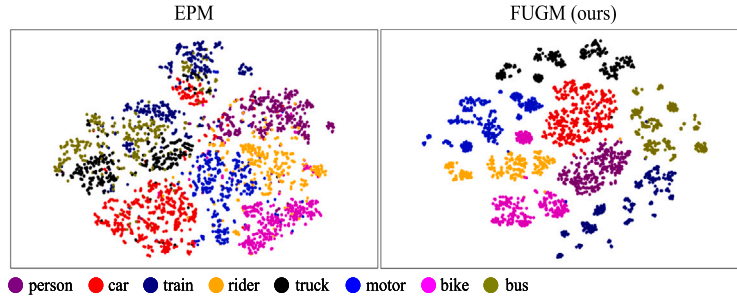


Fig. 4. For each category, we randomly sample object features (marked as circles) inside bounding boxes in the source domain and target domain equally.

Table 7

Comparison of hyperparameter settings λ_1 and λ_2 on KITTI $\rightarrow$ Cityscapes.

λ_1	λ_2	AP _{0.5:0.95}	AP ₅₀	AP _{0.75}
0.01	0.1	25.2	49.8	23.5
0.01	0.2	24.4	48.9	22.5
0.01	0.05	25.7	49.5	24.0
0.1	0.1	24.7	49.3	22.4
0.005	0.1	25.6	49.9	24.5

truth, as shown in Fig. 3. Compared to EPM (Hsu et al., 2020), both SIGMA and FUGM can reduce missing errors, such as the person and cars in the first row. In addition, FUGM can reduce false targets, for example, backgrounds are recognized as trains and trucks by SIGMA, due to the elimination of spurious matching with background features.

4.6.2. Feature T-SNE visualization

We present category-specific feature distributions in Fig. 4 to validate the effectiveness of our model. We randomly sample the same number of object features for each class in both domains. It can be intuitively observed that FUGM can clearly separate features of even similar categories (person, rider, bike), demonstrating the effectiveness of FUGM.

5. Discussion

5.1. Parameter sensitivity

As shown in Table 7, we conduct some experiments for analyzing how loss weight terms λ_1 and λ_2 , which separately control the degree of influence of the UGM and node classification. In our implementation, λ_1 and λ_2 are set to 0.01 and 0.1, respectively, for significant adaptation.

5.2. Failure case analysis

We have shown some examples of failures in Fig. 5, and through the analysis of these cases, we can obtain the following results. (1) Although many large objects have accurate bound and can obtain correct classification results, they are prone to repeated labeling, such as the train in the first row being repeatedly labeled. (2) In complex environments, our model may incorrectly predict some targets, such as the people in the second, third, and fourth rows. This may be due to our model overfitting to certain objects with similar appearances. (3) For some indistinguishable small objects, our model may not be able to classify them correctly, such as the person riding a bicycle in the fourth row, and our model may only be able to recognize the bicycle or person among them, making it difficult to accurately predict the category. Based on the above observations, we will continue to propose methods to try to solve these problems, such as using more robust universe graph generation method and introducing graph Multiview fusion methods.



Fig. 5. (a) Ground-truth and some (b) failure object detection of the proposed FUGM.

5.3. Graph matching visualization

In this section, we present the visualization results of our partial matching and full matching methods in Fig. 6, where the points are generated by saving the sampled point positions during the training process. During training, we set the batch size to 2 (one image from the source domain and one from the target domain). The circular points represent the sampled point positions, with different colors indicating different categories. The category labels in the source domain are derived from ground truth, while pseudo-labels are used in the target domain. Straight lines represent matched pairs of points predicted by our method, and the illusion nodes supplemented by GMB are shown in the lower-left and upper-left corners of the source and target domains, respectively. The yellow circle represents the mismatching with background, dashed yellow circle represents the non-existence of mismatches in the same region, dashed blue circle stands for the correct matching with backgrounds. From the visualization results, several observations can be made: (1) Our partial matching can suppress the spurious matching particularly between background and other foreground nodes. (2) Our partial matching helps model to focus on the right matching within the same category, which is necessary for fine-grained class-conditional distribution alignment.

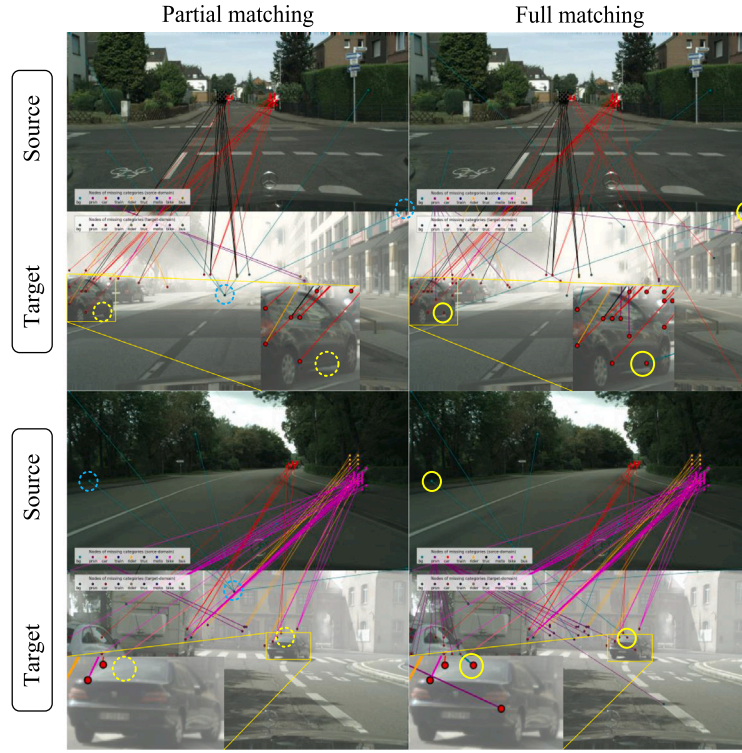


Fig. 6. Visualization of the comparisons between partial matching and full matching. The hallucination nodes in the higher-left and lower-left in source and target domain, respectively.

5.4. Extension to universal DAOD

The critical issues of the universal domain adaptive object detection involve the global-level private category alignment and the domain-shared category domain alignment (Zheng, Wu, Li, & Chen, 2025). Although we consider the DAOD scenarios in this work where the label space of the target domain is consistent with the source domain, our method can be easily extended to the universal domain adaptation. This is because our method mainly focuses on domain-shared category domain alignment by graph modeling, we can further solve the domain private classes through the constraint of aggregation of domain-private category centroids in feature and probability spaces, which can mitigate negative transfer. In solving such real-world scenario, it is important to calculate alignment weight for both of global-level and instance-level domain-shared category domain alignment, which can make use of our matching score matrix to generate weights for domain discriminator. With these modifications for the universal domain adaptation, we will focus on the extending robust graph-based paradigm to the complex real-world scenario in the future.

6. Conclusion

In this paper, we propose Foreground-aware Universe Graph Matching (FUGM) framework for DAOD, which solves the application problem of graph modeling and matching in DAOD. Specifically, it adopts a virtual universe graph based Collaborative Graph Reasoning (CGR) to model comprehensive knowledge concerning both domains and provides a filter to distinguish node category. Then, it leverages Universe Graph Matching (UGM) to encourage instance nodes to match anchor nodes and remove unreliable matching with absorbing nodes. The resulting matching model enables end-to-end learning especially for universe anchor node affinity enhancement with gathering foreground instance nodes together, which benefits the cross-domain semantics space alignment. Extensive experiments verify that FUGM outperforms existing works.

CRediT authorship contribution statement

Chao Wen: Writing – original draft, Conceptualization, Project administration. **Yongbo Wang:** Software, Visualization, Methodology. **Yuhua Qian:** Writing – review & editing, Funding acquisition, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Robustness in different noisy rates and lighting conditions

A.1. Datasets

A.1.1. Pascal VOC → Clipart(Watercolor)

The Pascal VOC dataset, comprising VOC 2007 and VOC 2012, is a real-world benchmark featured with 16,551 images across 20 common object categories for training. The Clipart and Watercolor datasets consist of artistic images collected from online sources. The Clipart dataset contains 1000 images annotated with the same 20 categories as Pascal VOC, while the Watercolor dataset includes 2000 images annotated for only 6 categories. In the Pascal VOC → Clipart setting, all Clipart images are used as the target domain for both training and evaluation. For the Pascal VOC → Watercolor setting, 1000 Watercolor images are allocated for training and the remaining 1000 images are used for testing.

Table 8Results of Pascal VOC and Noised Pascal VOC with different NRs $\rightarrow$ Clipart with Resnet101 backbone.

NR (%)	Method	aero	bicycle	bird	boat	bottle	bus	car	cat	chair	cow	table	dog	hrs	bike	prsn	plnt	sheep	sofa	train	tv	mAP
0	CP (Wang, Ma et al., 2019)	30.3	49.2	29.8	33.2	34.1	45.8	41.1	9.7	35.8	50.7	23.6	14.4	31.7	36.9	54.6	45.8	18.6	29.9	44.8	43.5	35.2
	GCE (Zhang & Sabuncu, 2018)	31.9	53.2	27.9	25.8	31.0	41.9	39.3	4.3	34.5	46.7	18.1	18.4	30.2	39.1	55.0	44.1	18.1	21.1	43.2	40.7	33.2
	NLTE (Liu et al., 2022)	39.1	50.3	33.6	34.7	35.0	40.5	44.2	5.9	36.8	45.8	23.1	17.3	31.8	39.5	60.7	45.4	17.9	28.4	49.0	51.3	36.5
	SIGMA (Li, Liu et al., 2022)	40.1	55.4	37.4	31.1	54.9	54.9	46.6	23.0	44.7	65.6	23.0	22.0	42.8	55.6	67.2	55.2	32.9	40.8	45.0	58.6	44.5
	FUGM (ours)	43.6	69.3	42.1	37.8	55.6	63.5	33.7	32.6	47.3	73.0	35.0	26.5	33.6	83.6	64.3	53.2	36.3	37.6	43.2	43.9	47.8
20	CP (Wang, Ma et al., 2019)	29.3	39.6	29.1	28.0	29.4	34.2	42.4	3.9	35.0	39.4	21.2	12.5	32.2	38.9	57.2	43.0	18.6	27.9	40.2	45.0	32.3
	GCE (Zhang & Sabuncu, 2018)	24.0	42.2	32.4	29.4	31.5	45.5	39.9	6.7	36.5	38.0	16.7	15.3	30.4	37.9	53.6	44.1	13.5	24.6	46.9	43.7	32.6
	NLTE (Liu et al., 2022)	33.1	47.5	35.5	28.2	33.7	53.8	43.8	4.2	34.2	48.4	19.3	14.6	29.7	47.2	57.1	42.5	17.7	27.7	40.0	44.5	35.1
	SIGMA (Li, Liu et al., 2022)	42.1	37.1	37.2	34.9	54.4	56.1	47.1	26.8	40.8	62.4	20.1	27.2	39.1	56.3	64.5	52.4	23.6	40.6	52.9	61.2	43.8
	FUGM (ours)	46.3	31.4	39.4	33.3	52.5	62.6	45.2	32.3	41.0	64.4	22.0	26.7	42.7	57.3	66.7	54.7	29.6	41.0	54.8	55.9	45.0
40	CP (Wang, Ma et al., 2019)	24.0	40.9	31.1	22.0	31.2	33.0	40.8	4.4	34.6	36.8	18.5	13.8	29.6	41.3	51.7	38.6	14.2	27.8	26.0	37.8	29.9
	GCE (Zhang & Sabuncu, 2018)	27.0	36.2	31.1	26.0	33.8	42.9	41.2	2.6	37.0	45.3	19.0	17.2	33.0	42.9	54.6	45.7	17.6	21.4	41.9	49.1	33.3
	NLTE (Liu et al., 2022)	32.8	45.5	30.8	29.8	35.7	43.2	43.0	6.4	32.7	45.9	19.8	10.8	31.1	43.4	56.4	43.3	19.6	24.8	42.5	43.9	34.1
	SIGMA (Li, Liu et al., 2022)	42.1	32.3	37.4	36.3	50.5	64.8	45.0	28.2	41.0	64.4	22.6	23.4	36.8	51.3	63.6	50.3	25.1	40.9	53.1	61.8	43.6
	FUGM (ours)	45.4	34.9	38.6	33.6	54.5	59.0	47.4	25.4	32.4	66.7	21.9	26.4	41.8	53.5	66.5	52.8	23.1	41.1	53.8	59.2	44.4

A.1.2. BDD100K Day $\rightarrow$ Night

BDD100K is an urban street scene dataset that can be divided into daytime and nighttime subsets based on the “timeofday” attribute. It can be adopted to evaluate the object detection performance under well-lit and low-light conditions, respectively. Each subset includes over 30,000 images. Nighttime object detection is challenging, so BDD100K provides an effective benchmark for evaluating domain adaptation performance across various lighting conditions.

A.2. Comparisons in different noisy rates

We list the results of using Pascal VOC and Noisy Pascal VOC with different noisy rates as the source domain, and Clipart1k as the target domain in Table 8. It is shown that the performances of the detectors drop consistently as the noisy rate (NR) increases. The performance of CP (Wang, Ma et al., 2019) severely drops by 2.9% and 5.3% at 20% and 40% NRs, compared with its performance under clean scenario. With generalized cross entropy loss method GCE (Zhang & Sabuncu, 2018), the detector performs better than CP under noisy settings, but they significantly deteriorate the performance of the detector under clean scenario, causing the mAP drops by 2.0%. With graph based method NLTE (Liu, Li, Yang, Li, & Yuan, 2022), SIGMA and FUGM, the detectors perform not only achieves robust performance under different NRs, but also guarantee the performance of the domain adaptive detector when the source annotations are clean. It is evident that the proposed FUGM outperforms over other methods in various NRs.

We also list the experimental results of using Pascal VOC and Noisy Pascal VOC with different NRs as the source domain, and Watercolor2k as the target domain in Table 9. It is shown that CP, GCE and NLTE methods perform bad, which indicates that they could suffer from the background interference and the lack of significant within-class variance modeling. As the noise rate increases from 20% to 40%, adopting FUGM consistently improves the mAP by 2.9% and 2.9% over SIGMA, respectively. The results evidently suggest that FUGM efficiently boosts the robustness of domain adaptive object detectors.

A.3. Comparisons in different lighting conditions

We examine our method on real driving scenes and evaluating the domain adaptation performance on nighttime images. The comparison results of this experiment are shown on Table 10. The results show that our method achieves the highest performance 45.4%. Compared with adversarial learning based method AT (Li, Dai et al., 2022) (38.5%), student-teacher based method UMT (Deng et al., 2021) (36.2%) and graph matching based method SIGMA (43.9%), our FUGM method has improved by 6.9%, 9.2% and 1.5%, respectively. This demonstrates the robustness of FUGM under low-light conditions.

Appendix B. Robustness in different background node variances

In the proposed framework, our aim is to improve the robustness of cross-domain graph matching rather than directly distinguish the background nodes for better DAOD. In order to verify the effectiveness of our partial graph matching framework for complex background node features with high variance, we have considered the multi-class DOAD task and implement FUGM in the Cityscapes $\rightarrow$ Foggy Cityscapes (C $\rightarrow$ F) benchmark. To further investigate how well our model distinguishes these background nodes with high variance, we tested the model performance under different variances. Specifically, we select the test images of background node features with the top 20% and the bottom 20% variances.

We study the impact of Absorbing Node Removal (ANR) on the performance, where w/o ANR stands for the full matching, otherwise it is FUGM. As is shown in Table 11, our FUGM performance slightly drops 1.1% (mAP) at the high variance while the full matching based method drops 2.0%. Our FUGM outperforms the full matching based method by 1.9% when dealing with the background nodes of high variance features. Therefore, our FUGM is more robust on handling the background nodes that might be difficult to distinguish. We also provide the image examples of high and low variance are selected for testing in Fig. 7. The top row shows the three selected image examples of background node features with high variance, and the bottom row shows the three selected image examples of background node features with low variance. The images in top row contain more semantically different backgrounds, such as traffic signs, buildings, roads, and traffic

Table 9

Results of Pascal VOC and Noised Pascal VOC with different NRs → Watercolor with Resnet101 backbone.

NR (%)	Method	bicycle	bird	car	cat	dog	prsn	mAP
0	CP (Wang, Ma et al., 2019)	67.1	39.1	34.5	27.2	22.9	45.3	39.4
	GCE (Zhang & Sabuncu, 2018)	67.3	37.0	39.4	21.9	21.3	46.4	38.9
	NLTE (Liu et al., 2022)	73.7	36.9	39.9	26.8	22.6	45.3	40.9
	SIGMA (Li, Liu et al., 2022)	76.3	52.9	56.9	34.7	38.5	69.1	54.8
	FUGM (ours)	83.5	56.1	59.0	33.9	40.9	73.7	57.8
20	CP (Wang, Ma et al., 2019)	72.0	36.5	21.3	18.3	21.1	41.5	35.1
	GCE (Zhang & Sabuncu, 2018)	62.7	42.5	40.1	26.2	18.8	44.9	39.2
	NLTE (Liu et al., 2022)	73.7	37.1	35.3	28.1	21.2	44.5	40.0
	SIGMA (Li, Liu et al., 2022)	78.6	53.8	49.0	38.6	26.6	69.0	52.6
	FUGM (ours)	76.4	52.5	56.6	41.5	35.6	70.3	55.5
40	CP (Wang, Ma et al., 2019)	66.0	36.6	17.8	24.0	18.2	39.8	33.7
	GCE (Zhang & Sabuncu, 2018)	64.3	40.0	34.7	21.3	19.0	43.8	37.2
	NLTE (Liu et al., 2022)	75.7	37.2	32.5	22.6	24.3	43.1	39.2
	SIGMA (Li, Liu et al., 2022)	80.3	51.4	53.4	32.1	26.0	66.4	51.6
	FUGM (ours)	78.0	50.8	54.0	37.6	26.8	69.7	54.5

Table 10

BDD100K Day → Night tasks with Resnet50 backbone.

Method	people	rider	car	truck	bus	motor	bicycle	traffic light	traffic sign	mAP
Da-Faster (Chen et al., 2018)	30.3	37.4	56.5	25.3	35.8	21.5	35.8	25.8	44.5	31.3
UMT (Deng et al., 2021)	46.5	26.1	46.8	44.0	46.3	28.2	40.2	31.6	52.7	36.2
TDD (He et al., 2022)	43.1	20.7	68.4	33.3	35.6	16.5	25.9	43.1	59.5	34.6
AT (Li, Dai et al., 2022)	42.3	30.4	60.8	48.9	52.1	34.5	42.7	29.1	43.9	38.5
SADAF (Chen, Wang et al., 2021)	33.0	35.1	67.7	24.3	38.6	33.0	49.7	51.5	61.4	38.4
SIGMA (Li, Liu et al., 2022)	56.0	24.7	72.3	51.7	51.4	27.9	38.8	42.7	61.7	43.9
FUGM (ours)	56.3	32.8	72.3	51.2	49.4	33.8	42.6	52.7	63.0	45.4

**Fig. 7.** The selected image examples of background node features with high and low variance.**Table 11**

The ablation results on different background node feature variances.

Method	Average variance	$AP_{0.5:0.95}$	$AP_{0.5}$	$AP_{0.75}$
w/o ANR	Top 20%	24.3	43.7	25.6
	Bottom 20%	27.0	45.7	24.8
	Top 20%	26.3	45.6	26.0
FUGM	Bottom 20%	28.9	46.7	29.3

lights, while the background of the images in bottom row is more simple and mainly contains trees and roads.

Appendix C. Computational complexity

We provide the analysis of computational costs of the backbone, FCOS detector and graph module have been calculated. The graph

module includes graph construction with subgraph-level fusion, collaborative graph reasoning (CGR) and universe graph matching (UGM). VGG-16 is adopted as the backbone. Note that the computational complexity analyzed for training stage, because the graph module is not required for inference stage.

As is shown in Table 12, our model requires a total computation of 2570.1GFLOPs, where the backbone accounts for 2211.3GFLOPs, the FCOS detector for 357.4GFLOPs and the graph module for 1.4GFLOPs. It is evident that the graph module requires only 0.06% of the overall computation cost. As a result, the computational cost of the graph model involving CGR and UGM is considerably low. This demonstrates that our graph module is more computationally efficient than other modules based on CNN like backbone and detector. When we optimize the model training efficiency for real scenarios, it is necessary for first considering efficient backbone and detector so as to remarkably improve the whole model training efficiency.

Table 12
Analysis of computational complexity with VGG-16 as backbone.

	Backbone (VGG-16)	FCOS	Graph module (CGR, UGM)	Total
Computational cost (GFLOPs)	2211.3	357.4	1.4	2570.1
Percentage of total cost	86.04%	13.90%	0.06%	100%

References

- Arruda, V. F., Paixao, T. M., Berriel, R. F., De Souza, A. F., Badue, C., Sebe, N., et al. (2019). Cross-domain car detection using unsupervised image-to-image translation: From day to night. In *International joint conference on neural networks* (pp. 1–8). IEEE.
- Bai, Y., Liu, C., Yang, R., & Li, X. (2024). Misalignment-resistant domain adaptive learning for one-stage object detection. *Knowledge-Based Systems*, Article 112605.
- Carion, N., Massa, F., Synnaeve, G., Usunier, N., Kirillov, A., & Zagoruyko, S. (2020). End-to-end object detection with transformers. In *European conference on computer vision* (pp. 213–229). Springer.
- Chang, X., Ren, P., Xu, P., Li, Z., Chen, X., & Hauptmann, A. (2021). A comprehensive survey of scene graphs: Generation and application. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 45(1), 1–26.
- Chen, J., Deng, W., Peng, B., Liu, T., Wei, Y., & Liu, L. (2023). Variational information bottleneck for cross domain object detection. In *IEEE international conference on multimedia and expo* (pp. 2231–2236). IEEE.
- Chen, Y., Li, W., Sakaridis, C., Dai, D., & Van Gool, L. (2018). Domain adaptive faster r-cnn for object detection in the wild. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition* (pp. 3339–3348).
- Chen, C., Li, J., Zheng, Z., Huang, Y., Ding, X., & Yu, Y. (2021). Dual bipartite graph learning: A general approach for domain adaptive object detection. In *Proceedings of the IEEE/CVF international conference on computer vision* (pp. 2703–2712).
- Chen, C., Li, J., Zhou, H.-Y., Han, X., Huang, Y., Ding, X., et al. (2022). Relation matters: Foreground-aware graph-based relational reasoning for domain adaptive object detection. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 45(3), 3677–3694.
- Chen, Y., Wang, H., Li, W., Sakaridis, C., Dai, D., & Van Gool, L. (2021). Scale-aware domain adaptive faster r-cnn. *International Journal of Computer Vision*, 129(7), 2223–2243.
- Chen, Z.-M., Wei, X.-S., Wang, P., & Guo, Y. (2021). Learning graph convolutional networks for multi-label recognition and applications. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 45(6), 6969–6983.
- Chen, C., Zheng, Z., Huang, Y., Ding, X., & Yu, Y. (2021). I3net: Implicit instance-invariant network for adapting one-stage object detectors. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition* (pp. 12576–12585).
- Cordts, M., Omran, M., Ramos, S., Rehfeld, T., Enzweiler, M., Benenson, R., et al. (2016). The cityscapes dataset for semantic urban scene understanding. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition* (pp. 3213–3223).
- Deng, J., Li, W., Chen, Y., & Duan, L. (2021). Unbiased mean teacher for cross-domain object detection. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition* (pp. 4091–4101).
- Fu, K., Luo, J., Luo, X., Liu, S., Zhang, C., & Wang, M. (2022). Robust point cloud registration framework based on deep graph matching. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 45(5), 6183–6195.
- Ganin, Y., Ustinova, E., Ajakan, H., Germain, P., Larochelle, H., Laviolette, F., et al. (2016). Domain-adversarial training of neural networks. *Journal of Machine Learning Research*, 17(59), 1–35.
- Gao, C., Liu, C., Dun, Y., & Qian, X. (2023). CSDA: Learning category-scale joint feature for domain adaptive object detection. In *Proceedings of the IEEE/CVF international conference on computer vision* (pp. 11421–11430).
- Gao, Q., Wang, F., Xue, N., Yu, J.-G., & Xia, G.-S. (2021). Deep graph matching under quadratic constraint. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition* (pp. 5069–5078).
- Geiger, A., Lenz, P., & Urtasun, R. (2012). Are we ready for autonomous driving? the kitti vision benchmark suite. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition* (pp. 3354–3361). IEEE.
- Girshick, R. (2015). Fast r-cnn. In *Proceedings of the IEEE international conference on computer vision* (pp. 1440–1448).
- Hartmanis, J. (1982). Computers and intractability: a guide to the theory of np-completeness (michael r. Garey and david s. Johnson). *Siam Review*, 24(1), 90.
- He, M., Wang, Y., Wu, J., Wang, Y., Li, H., Li, B., et al. (2022). Cross domain object detection by target-perceived dual branch distillation. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition* (pp. 9570–9580).
- He, K., Zhang, X., Ren, S., & Sun, J. (2016). Deep residual learning for image recognition. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition* (pp. 770–778).
- He, Z., Zhang, L., Yang, Y., & Gao, X. (2021). Partial alignment for object detection in the wild. *IEEE Transactions on Circuits and Systems for Video Technology*, 32(8), 5238–5251.
- Hsu, C.-C., Tsai, Y.-H., Lin, Y.-Y., & Yang, M.-H. (2020). Every pixel matters: Center-aware feature alignment for domain adaptive object detector. In *European conference on computer vision* (pp. 733–748). Springer.
- Ji, T., Wu, Y., & Lan, M. (2019). Graph-based dependency parsing with graph neural networks. In *Proceedings of the 57th annual meeting of the association for computational linguistics* (pp. 2475–2485).
- Johnson-Roberson, M., Barto, C., Mehta, R., Sridhar, S. N., Rosaen, K., & Vasudevan, R. (2017). Driving in the matrix: Can virtual worlds replace human-generated annotations for real world tasks? In *IEEE international conference on robotics and automation* (pp. 746–753). IEEE.
- Ke, J., He, L., Han, B., Li, J., Wang, D., & Gao, X. (2024). Vldadaptor: Domain adaptive object detection with vision-language model distillation. *IEEE Transactions on Multimedia*.
- Li, Y.-J., Dai, X., Ma, C.-Y., Liu, Y.-C., Chen, K., Wu, B., et al. (2022). Cross-domain adaptive teacher for object detection. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition* (pp. 7581–7590).
- Li, P., He, Y., Yu, F. R., Song, P., Yin, D., & Zhou, G. (2023). IGG: Improved graph generation for domain adaptive object detection. In *Proceedings of the 31st ACM international conference on multimedia* (pp. 1314–1324).
- Li, W., Liu, X., & Yuan, Y. (2022). Sigma: Semantic-complete graph matching for domain adaptive object detection. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition* (pp. 5291–5300).
- Li, W., Liu, X., & Yuan, Y. (2023). Scan++: Enhanced semantic conditioned adaptation for domain adaptive object detection. *IEEE Transactions on Multimedia*, 25, 7051–7061.
- Li, W., Liu, X., & Yuan, Y. (2023c). Sigma++: Improved semantic-complete graph matching for domain adaptive object detection. *IEEE Transactions on Pattern Analysis and Machine Intelligence*.
- Lin, C., Yuan, Z., Zhao, S., Sun, P., Wang, C., & Cai, J. (2021). Domain-invariant disentangled network for generalizable object detection. In *Proceedings of the IEEE/CVF international conference on computer vision* (pp. 8771–8780).
- Liu, X., Li, W., Yang, Q., Li, B., & Yuan, Y. (2022). Towards robust adaptive object detection under noisy annotations. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition* (pp. 14207–14216).
- Liu, Y., Wang, J., Huang, C., Wang, Y., & Xu, Y. (2023). CIGAR: Cross-modality graph reasoning for domain adaptive object detection. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition* (pp. 23776–23786).
- Liu, Y., Wang, J., Wang, W., Hu, Y., Wang, Y., & Xu, Y. (2024). CRADA: Cross domain object detection with cyclic reconstruction and decoupling adaptation. *IEEE Transactions on Multimedia*, 26, 6250–6261.
- Liu, Y., Wang, J., Xiao, L., Liu, C., Wu, Z., & Xu, Y. (2023). Foregroundness-aware task disentanglement and self-paced curriculum learning for domain adaptive object detection. *IEEE Transactions on Neural Networks and Learning Systems*.
- Liu, F., Zhang, X., Wan, F., Ji, X., & Ye, Q. (2021). Domain contrast for domain adaptive object detection. *IEEE Transactions on Circuits and Systems for Video Technology*, 32(12), 8227–8237.
- Loiola, E. M., De Abreu, N. M. M., Boaventura-Netto, P. O., Hahn, P., & Querido, T. (2007). A survey for the quadratic assignment problem. *European Journal of Operational Research*, 176(2), 657–690.
- Menke, M., Wenzel, T., & Schwung, A. (2024). Awada: Foreground-focused adversarial learning for cross-domain object detection. *Computer Vision and Image Understanding*, Article 104153.
- Munir, M. A., Khan, M. H., Sarfraz, M., & Ali, M. (2021). Ssal: Synergizing between self-training and adversarial learning for domain adaptive object detection. *Advances in Neural Information Processing Systems*, 34, 22770–22782.
- Paszke, A., Gross, S., Massa, F., Lerer, A., Bradbury, J., Chanan, G., et al. (2019). Pytorch: An imperative style, high-performance deep learning library. *Advances in Neural Information Processing Systems*, 32.
- Pei, H., Wei, B., Chang, K. C. C., Lei, Y., & Yang, B. (2020). Geom-gcn: geometric graph convolutional networks. In *8th international conference on learning representations*.
- Ramamonjison, R., Banitalebi-Dehkordi, A., Kang, X., Bai, X., & Zhang, Y. (2021). Simrod: A simple adaptation method for robust object detection. In *Proceedings of the IEEE/CVF international conference on computer vision* (pp. 3570–3579).
- Redmon, J., & Farhadi, A. (2018). Yolov3: An incremental improvement. *arXiv preprint arXiv:1804.02767*.
- Rezaeianaran, F., Shetty, R., Aljundi, R., Reino, D. O., Zhang, S., & Schiele, B. (2021). Seeking similarities over differences: Similarity-based domain alignment for adaptive object detection. In *Proceedings of the IEEE/CVF international conference on computer vision* (pp. 9204–9213).
- Saito, K., Ushiku, Y., Harada, T., & Saenko, K. (2019). Strong-weak distribution alignment for adaptive object detection. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition* (pp. 6956–6965).

- Sakarisid, C., Dai, D., & Van Gool, L. (2018). Semantic foggy scene understanding with synthetic data. *International Journal of Computer Vision*, 126, 973–992.
- Shi, W., & Rajkumar, R. (2020). Point-gnn: Graph neural network for 3d object detection in a point cloud. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition* (pp. 1711–1719).
- Simonyan, K., & Zisserman, A. (2015). Very deep convolutional networks for large-scale image recognition. In *International conference on learning representations, international conference on learning representations*.
- Tian, Z., Shen, C., Chen, H., & He, T. (2019). FCOS: Fully convolutional one-stage object detection. In *Proceedings of the IEEE/CVF international conference on computer vision* (pp. 9627–9636).
- Tian, K., Zhang, C., Wang, Y., Xiang, S., & Pan, C. (2021). Knowledge mining and transferring for domain adaptive object detection. In *Proceedings of the IEEE/CVF international conference on computer vision* (pp. 9133–9142).
- Torralba, A., & Efros, A. A. (2011). Unbiased look at dataset bias. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition* (pp. 1521–1528). IEEE.
- Tzeng, E., Hoffman, J., Saenko, K., & Darrell, T. (2017). Adversarial discriminative domain adaptation. In *Proceedings of the IEEE conference on computer vision and pattern recognition* (pp. 7167–7176).
- Vs, V., Gupta, V., Oza, P., Sindagi, V. A., & Patel, V. M. (2021). Mega-cda: Memory guided attention for category-aware unsupervised domain adaptive object detection. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition* (pp. 4516–4526).
- Wang, Z., Luo, Y., Huang, Z., & Baktashmotlagh, M. (2020). Prototype-matching graph network for heterogeneous domain adaptation. In *Proceedings of the 28th ACM international conference on multimedia* (pp. 2104–2112).
- Wang, Y., Ma, X., Chen, Z., Luo, Y., Yi, J., & Bailey, J. (2019). Symmetric cross entropy for robust learning with noisy labels. In *Proceedings of the IEEE/CVF international conference on computer vision* (pp. 322–330).
- Wang, R., Yan, J., & Yang, X. (2019). Learning combinatorial embedding networks for deep graph matching. In *Proceedings of the IEEE/CVF international conference on computer vision* (pp. 3056–3065).
- Wang, Y., Yasunaga, M., Ren, H., Wada, S., & Leskovec, J. (2023). Vqa-gnn: Reasoning with multimodal knowledge via graph neural networks for visual question answering. In *Proceedings of the IEEE/CVF international conference on computer vision* (pp. 21582–21592).
- Wang, Y., Zhang, R., Zhang, S., Li, M., Xia, Y., Zhang, X., et al. (2021). Domain-specific suppression for adaptive object detection. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition* (pp. 9603–9612).
- Weng, X., Wang, Y., Man, Y., & Kitani, K. M. (2020). Gnn3dmot: Graph neural network for 3d multi-object tracking with 2d-3d multi-feature learning. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition* (pp. 6499–6508).
- Wu, L., Chen, Y., Ji, H., & Liu, B. (2021). Deep learning on graphs for natural language processing. In *Proceedings of the 44th international ACM SIGIR conference on research and development in information retrieval* (pp. 2651–2653).
- Xi, S., Liu, Y., Lu, H., Ding, Y., & Ding, H. (2025). GDRIVE: Adaptive object detection in autonomous vehicles via graph-based feature learning. In *ICASSP 2025-2025 IEEE international conference on acoustics, speech and signal processing* (pp. 1–5). IEEE.
- Xu, M., Wang, H., Ni, B., Tian, Q., & Zhang, W. (2020). Cross-domain detection via graph-induced prototype alignment. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition* (pp. 12355–12364).
- Xu, S., Zhang, H., Xu, X., Hu, X., Xu, Y., Dai, L., et al. (2022). Representative feature alignment for adaptive object detection. *IEEE Transactions on Circuits and Systems for Video Technology*, 33(2), 689–700.
- Yan, J., Yin, X.-C., Lin, W., Deng, C., Zha, H., & Yang, X. (2016). A short survey of recent advances in graph matching. In *Proceedings of the 2016 ACM on international conference on multimedia retrieval* (pp. 167–174).
- Yang, X., Liu, Z.-Y., & Qiao, H. (2020). A continuation method for graph matching based feature correspondence. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 42(08), 1809–1822.
- Yoo, J., Chung, I., & Kwak, N. (2022). Unsupervised domain adaptation for one-stage object detector using offsets to bounding box. In *European conference on computer vision* (pp. 691–708). Springer.
- Yuan, J., Ma, X., Chen, D., Kuang, K., Wu, F., & Lin, L. (2023). Domain-specific bias filtering for single labeled domain generalization. *International Journal of Computer Vision*, 131(2), 552–571.
- Zhang, Z., & Sabuncu, M. (2018). Generalized cross entropy loss for training deep neural networks with noisy labels. *Advances in Neural Information Processing Systems*, 31.
- Zhang, Y., Wang, Z., & Mao, Y. (2021). Rpn prototype alignment for domain adaptive object detector. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition* (pp. 12425–12434).
- Zhang, B., Xiao, J., Jiao, J., Wei, Y., & Zhao, Y. (2021). Affinity attention graph neural network for weakly supervised semantic segmentation. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 44(11), 8082–8096.
- Zhang, L., Zhang, B., Shi, B., Fan, J., & Chen, T. (2024). Few-shot cross-domain object detection with instance-level prototype-based meta-learning. *IEEE Transactions on Circuits and Systems for Video Technology*, 34(10), 9078–9089.
- Zhao, L., & Wang, L. (2022). Task-specific inconsistency alignment for domain adaptive object detection. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition* (pp. 14217–14226).
- Zheng, Y., Wu, J., Li, W., & Chen, Z. (2025). Universal domain adaptive object detection via dual probabilistic alignment. *arXiv preprint arXiv:2412.11443*.
- Zhou, W., Du, D., Zhang, L., Luo, T., & Wu, Y. (2022). Multi-granularity alignment domain adaptation for object detection. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition* (pp. 9581–9590).
- Zhou, X., Zhu, M., & Daniilidis, K. (2015). Multi-image matching via fast alternating minimization. In *Proceedings of the IEEE/CVF international conference on computer vision* (pp. 4032–4040).
- Zhu, Y., Liu, Y., Wang, C., Wang, S., & Lu, M. (2023). Intermediate domain based meta learning framework for adaptive object detection. *IEEE Transactions on Circuits and Systems for Video Technology*, 34(7), 5255–5265.